

GCSE algebra - Set 2

TMUA practice questions by topic. Specification M4.1-M4.8, M4.18, M4.19. 3 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The distance-time graph of a journey is a straight line from (0, 0) to (2, 60), then horizontal to (3, 60), then a straight line to (5, 0) (time in hours, distance from home in km). What was the average speed for the whole journey?

A 0 km/h
B 12 km/h
C 24 km/h
D 20 km/h
E 30 km/h
F 40 km/h

- 2 Simplify $\frac{x^2 - 9}{(x + 3)(x - 4)} \times \frac{x^2 - 16}{x^2 - 3x}$.

A $\frac{x - 4}{x(x - 3)}$
B $\frac{x - 3}{x + 4}$
C $\frac{x + 3}{x - 4}$
D $\frac{x + 4}{x - 3}$
E $\frac{(x + 3)(x + 4)}{(x - 3)(x - 4)}$
F $\frac{(x - 3)(x - 4)}{x(x + 3)}$

- 3 The n th term of a sequence is $\frac{n^2 + 1}{2n}$. Which of the following statements is/are true?

I The 5th term is 2.6.
II Every term after the first is greater than the previous term.
III Some term of the sequence equals 10.

A none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III